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Advancing PFASs monitoring in food: from targeted SPE-LC-MS/MS to non-targeted QuEChERS-LC-HRMS approaches

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ABSTRACT

Per- and polyfluoroalkyl substances (PFASs) are increasingly recognized as emerging contaminants, with some classified as persistent organic pollutants (POPs) and four recently regulated in food in the EU. However, due to the large structural diversity of PFASs, comprehensive monitoring remains essential. This study aimed to develop and apply a suspect and non-targeted screening approach to identify PFASs beyond those conventionally monitored in foodstuffs. The methodology combined optimized QuEChERS sample preparation, LC-HRMS acquisition, and prioritization of fluorinated signals, applying both suspect screening (SS) and non-targeted screening (NTS) strategies. Compared to targeted solid-phase extraction (SPE) coupled with MS/MS (QqQ) acquisition, this approach detected a broader range of PFASs in various food samples. Notably, as one of the first studies to apply NTS—a method typically used in environmental analysis—to food, it demonstrated the ability to detect both known and previously unlisted PFASs, such as PFPrA in an egg sample. This expanded approach enhances exposure assessment and supports the implementation of HRMS-based strategies for regulatory control and risk assessment.

1. Introduction

Per- and poly-fluoroalkyl substances (PFASs) are man-made compounds widely used since the 1950s in many products, making them ubiquitous in the environment and a source of chronic exposure (Glüge et al., 2020; Schymanski et al., 2023). Compounds like PFOA, PFOS, and PFHxS are listed as persistent organic pollutants (POPs) under the Stockholm Convention due to their persistence, bioaccumulation, toxicity, and long-range transport (Zahm et al., 2024). Food is one of the main routes of non-occupational exposure, with some PFASs found in protein-rich foods due to their affinity for proteins (Abunada et al., 2020; Aker et al., 2023; Eze et al., 2024).

The European Food Safety Authority (EFSA) set a tolerable weekly intake (TWI) of 4.4 $ng.kg^{-1}$ body weight for the sum of PFOA, PFOS, PFNA, and PFHxS due to immune system effects (EFSA, 2020). This led to maximum levels in food, e.g., 1.7 $\mu g.kg^{-1}$ for eggs and up to 50 $\mu g.kg^{-1}$ for game offal (European Commission, 2022a). While only four PFASs are regulated, their number has risen from 4700 in 2018 (Wang et al., 2021) to over 8000 in 2022 (Kaufmann et al., 2022), with PubChem listing over 7 million (Schymanski et al., 2023). EFSA recommends improving analytical methods and expanding PFAS monitoring and regulation (EFSA, 2020; European Commission, 2022b).

In response, the scientific community has advanced methods to detect more PFASs by enhancing preparation, separation, and data processing. Though mainly applied to environmental samples, food—despite being a key exposure route—has been less studied. With increasing concern from health agencies calling for broader monitoring, these approaches aim to better characterize food contamination and improve exposure assessments. (EFSA, 2020; European Commission, 2022b).

Solid-phase extraction (SPE) is widely used for purifying PFASs in food samples when targeting specific compounds. Techniques include weak anion exchangers (WAX), hydrophilic-lipophilic balance (HLB) cartridges, or adsorbents like Graphitized Carbon Black (GCB), commonly paired with LC-MS/MS (Liquid Chromatography coupled to tandem Mass Spectrometry (MS^2)) analysis (Iannone et al., 2024; Pan et al., 2020). These methods have been applied in studies such as the French total diet study (Rivière et al., 2014) and analyses of various food matrices (Jaus et al., 2023; Meng et al., 2022).

The QuEChERS (Quick, Easy, Cheap, Effective, Rugged, Safe) method, initially for pesticides (Anastassiades et al., 2003), is now adapted for PFASs analysis in food (Table 1) (Iannone et al., 2024). It

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involves solvent extraction, typically using acetonitrile (ACN) acidified with formic acid (FA) or alternatives like methanol (MeOH) or methyl tert-butyl ether (MTBE) (Acosta-Dacal et al., 2020; Rivière et al., 2014; Taniyasu et al., 2003), followed by a salting-out step to separate phases, commonly using $MgSO_4$/NaCl (Genualdi et al., 2022; Kim et al., 2019; Roberts et al., 2023). The dispersive-SPE (d-SPE) step uses adsorbents like Primary Secondary Amine (PSA), GCB, octadecyl silica (C18) or $MgSO_4$ to remove interferences (i.e cholic acid, which can signal can interfere with some target compounds) and retain PFASs (Genualdi et al., 2022; Sadighara et al., 2024).

Filtration may be applied for complex matrices (Table 1), with SPE cartridges used if additional purification is needed, though this can reduce recoveries (Genualdi et al., 2022; Sadighara et al., 2024). QuEChERS offers a versatile, effective approach, critical for expanding PFASs extraction from diverse food matrices.

After non-selective sample preparation, selecting the appropriate analytical method is key to expanding the detection of PFASs through non-targeted analysis. Most studies extract PFASs from food samples using QuEChERS protocols and analyze them with LC-MS/MS for targeted quantification (Genualdi et al., 2022; Lacina et al., 2011; Sznajder-Katarzyńska et al., 2018). This method allows monitoring of up to 45 PFASs (Meng et al., 2022; Rivière et al., 2014) but struggles to identify certain compounds like PFBA and PFPeA, as they only have one ion transition with QQQ, which require high-resolution mass spectrometry (HRMS) (Genualdi et al., 2022). HRMS is increasingly used for wider screening, including suspect and non-targeted analyses (Knolhoff & Fisher, 2021). Qualitative analysis through MS/MS and data acquisition methods (DDA, DIA, AIF) helps minimize false detections but presents challenges for complex samples (Chu & Letcher, 2024; Pan et al., 2020).

Data processing is essential for detecting PFASs using mass spectrometry and prioritizing key signals in HRMS data. Common features include homologue series (CF2), diagnostic fragment ions (DFIs), neutral loss, and Kendrick Mass Defect (KMD) plots. Techniques like mass defect filtering and exact mass matching with suspect lists aid PFASs identification (Bugsel et al., 2023; Feng et al., 2024). Software tools such as FluoroMatch and Compound Discoverer are widely used for screening PFASs (Chu & Letcher, 2024; Cooper & Yang, 2024; Feng et al., 2024; Kaufmann et al., 2022; Knolhoff & Fisher, 2021), with other programs like FindPFAS (Zweigle et al., 2022) and PFAScreen (Zweigle et al., 2024) also reported.

In line with recent recommendations (EFSA, 2020; European Commission, 2022b), this work addresses a critical need to expand the spectrum of analyzable PFASs beyond the targeted ISO17025 method used in national monitoring plans, thereby responding to the growing demand from risk assessors and managers for more comprehensive surveillance and control strategies. These professionals have highlighted the limitations of current monitoring approaches, which primarily focus on a narrow range of PFASs compounds, and have called for methods that can detect a broader array of these contaminants to better assess environmental contamination and human exposure, particularly in foodstuffs (Government of France, 2024). To this and, a novel application of a QuEChERS extraction method was initially developed to target a finite set of 28 PFASs, along with 15 labelled standards, reflecting those typically monitored in routine methods. This sample preparation method was applied across various food matrices, including fish and milk, and further coupled with non-targeted HRMS analysis. Its performance was then evaluated in both SS and non-targeted screening (NTS) context for the characterization of food samples of animal origin, demonstrating the potential of this approach to advance current monitoring capabilities. By allowing the detection of both regulated and previously unmonitored PFAS compounds, this study directly supports the advancement of more comprehensive surveillance frameworks. Its results are especially valuable for regulatory bodies seeking to enhance exposure assessments and revise contaminant lists in response to emerging public health issues.

2. Materials and methods

2.1. Chemicals and reagents

Ammonium acetate, KOH tablets and ammonia 32 % (NH_4OH), and Sodium Chloride (NaCl) were obtained from Merck (Darmstadt, Germany). Methanol (MeOH) picograde was from Fisher Scientific, a brand of Thermo Fisher Scientific, ACN for LC-MS and glacial acetic acid (ACA) from Carlo-Erba Reagents (Rodano, Italy). Acetonitrile for LC-HRMS was obtained from Honeywell, and FA and supelclean ENVI-Carb SPE tube from Supelco (Sigma-Aldrich, Saint-Quentin Fallavier, France). HF Bond Elut – C18 (Oasis WAX) was obtained from Agilent Technologies (Santa Clara, CA, USA), and Magnesium Sulfate ($MgSO_4$) from ThermoScientific. Ultrapure water (>14 MΩ.cm) and water for LC-HRMS from VWR (Radnor, PA, USA) were used. The 0.22 µm nylon syringe filters were from Ait France (Cormeilles-en-Parisis, France). The C18 and GCB sorbents used in the d-SPE step of the QuEChERS sample preparation protocol were recovered from the SPE cartridges mentioned earlier.

The method was developed using 28 PFASs compounds plus their 15 labelled reference compounds. Standards of all selected PFASs (standards and internal standards) were purchased from Wellington

Table 1
Representative examples of the use of QuEChERS to extract PFASs from foods.

Matrice	Solvent extraction	Salting-out	Clean-up (d-SPE)	Filtration	Reference
First QuEChERS protocol					
Fruits and vegetables	ACN	MgSO4/NaCl (4:1, w/w)	$MgSO_4$/PSA	–	(Anastassiades et al., 2003)
Focus on PFASs extraction					
Fish	ACN	–	ENVI-Carb graphitized carbon	–	(Powley et al. 2008)
Fish and milk	ACN/H_2O/FA	MgSO4/NaCl (4:1, *w*/w)	$MgSO_4$/C18/GCB	0.2 µm polyvinylidene fluoride filter	(Lacina et al., 2011)
Milk and baby food	ACN/H_2O/FA	MgSO4/NaCl (4:1, w/w)	$MgSO_4$/C18	0.22 µm nylon filter	(Lankova et al. 2013)
Fruits and vegetables	ACN/FA	MgSO4/NaCl (4:1, w/w)	$MgSO_4$/ENV SPE Bulk	Paper filter	(Sznajder-Katarzyńska et al., 2018)
Food	ACN/H_2O/FA	MgSO4/NaCl (4:1, w/w)	$MgSO_4$/PSA/GCB	0.2 µm nylon filter	(Genualdi et al., 2022)
Fish, meat, and eggs	ACN/H_2O	MgSO4/NaCl (4:1, w/w)	$MgSO_4$/PSA/C18/CarbonX	–	(Taylor & Sapozhnikova, 2022)
Tomato, strawberry, and milk	ACN	NaCl	–	0.45 µm nylon filter	(Roberts et al., 2023)

Laboratories. A solution of all ($n = 28$) analytical standards containing the 4 regulated PFASs (European Commission, 2022a) but also the following 24 PFASs, called "PFASs mix", was prepared in methanol (0.1 ng.μL^{-1} each): perfluorobutanoic acid (PFBA), perfluoropentanoic acid (PFPeA), perfluorobutane sulfonic acid (PFBS), perfluorohexanoic acid (PFHxA), perfluoroheptanoic acid (PFHpA), perfluorohexane sulfonate (PFHxS), perfluorooctanoic acid (PFOA), perfluoroheptane sulfonic acid (PFHpS), perfluorooctane sulfonic acid (PFOS), perfluorononanoic acid (PFNA), perfluorodecanoic acid (PFDA), per-fluoroundecanoic acid (PFUnDA), perfluorodecane sulfonic acid (PFDS), perfluorododecanoic acid (PFDoDA), perfluorotridecanoic acid (PFTrDA), per-fluorotetradecanoic acid (PFTeDA), perfluorooctane sulfonamide (FOSA), 2-(*N*-methylperfluorooctanesulfonamido) acetic acid (MeFOSAA), 2-(N-ethylperfluorooctanesulfonamido) acetic acid (EtFOSAA), Perfluoropentanesulfonate (PFPeS), Perfluorononanesulfonate (PFNS), Perfluoroundecanesulfonate (PFUnDS), Perfluorododecanesulfonate (PFDoDS), Perfluorotridecanesulfonate (PFTrDS), Dodecafluoro-3H-4,8-dioxanonanoate (NaDONA), 2,3,3,3-Tetrafluoro-2-(1,1,2,2,3,3,3-hepta-fluoropropoxy) propanoic acid (GenX or HFPO-DA), 9-Chlorohexadeca-fluoro-3-oxanonane-1-sulfonate (9Cl-PF3ONS), 11-Chloroeicosafluoro-3-oxaundecane-1-sulfonate (11Cl-PF3OUdS).

A solution containing the following labelled PFASs used as internal standards (IS) ($n = 15$) was prepared in methanol (0.1 ng.μL^{-1} each): PFBA 13C4, PFPeA 13C3, PFBS 13C3, PFHxA 13C5, PFHpA13C4, PFHxS18O2, PFOA 13C4, PFOS 13C4, PFNA 13C5, PFDA 13C2, PFUnDA 13C2, PFDoDA13C2, FOSA 13C8, 1 d5-EtFOSAA, 1 d3-MeFOSAA. A solution of $13C_8$ PFOS was prepared in methanol (0.01 ng/μL), as an external standard.

2.2. Investigated samples

Food items of animal origin were collected in France in 2021 as part of a previous study (Godéré et al., 2024), aiming to represent typical consumption patterns in France. These items were selected based on the third French individual and national food consumption survey (INCA3), conducted by the French Agency for Food, Environmental, and Occupational Health and Safety (ANSES) in 2014–2015. A subset of 20 items from supermarkets ($n = 20$) was selected as follows: three eggs samples ($n = 3$), four seafood, shellfish, molluscs samples (cooked shrimps (n = 1), scallops (n = 1), raw mussels (n = 1) and cooked mussels (n = 1)), five fish samples (trout fillet (n = 1), back of cod (n = 1), alaska hake fillet (n = 1), smoked salmon (n = 1) and sardines (n = 1), five meat samples (butcher steak (beef) (n = 1), côtelettes (lamb) (n = 1), white ham with rind (pork) (n = 1), chipolata (pork) (n = 1) and thigh with skin (chicken) (n = 1)) and three dairy products samples (grated emmental (n = 1), goat's cheese bûche (n = 1) and plain cow's milk yoghurt (n = 1)). These samples were stored at −20 °C until analysis.

2.3. Sample preparation

2.3.1. Sample pretreatment

Samples were placed individually in petri dishes, which have been placed at −20 °C for at least 15 h, then freeze-dried during at least 36 h. Freeze-dried samples were homogenized and grounded to a powder, according to a previous work (Amziane et al., 2022), and stored at −20 °C.

2.3.2. Solid phase extraction (SPE)

Samples were extracted according to the extraction method reported in 2014 (Rivière et al., 2014), with minor modifications, and presented in Fig. 1. Infant milk was prepared as described on the milk powder pack: 150 mg milk powder with 1 mL ultrapure water (at room temperature) for 1 mL milk. In order to develop a protocol applicable to milk and fish, the amount of freeze-dried fish was also set at 150 mg. Briefly, 60 μL of the IS solution was added to 150 mg dry sample. Then, the samples were mixed, left in contact with the IS solution for 15 min, and 3 mL of 0.01 M potassium hydroxide methanoic (MeOH/KOH) solution were added. The mixture was stored overnight at room temperature. The supernatant was evaporated under nitrogen to 1 mL. After addition of 4 mL of ultrapure water, two 6 mL SPE cartridges (Oasis WAX and ENVI-Carb) purifications were successively performed. The extract was dried under nitrogen and reconstituted in 90 μL of MeOH ($13C_8$ PFOS)/H_2O (40/60; *v*/v), for LC-MS/MS and LC-HRMS analysis.

2.3.3. QuEChERS

The optimisation of QuEChERS extraction protocol is presented in

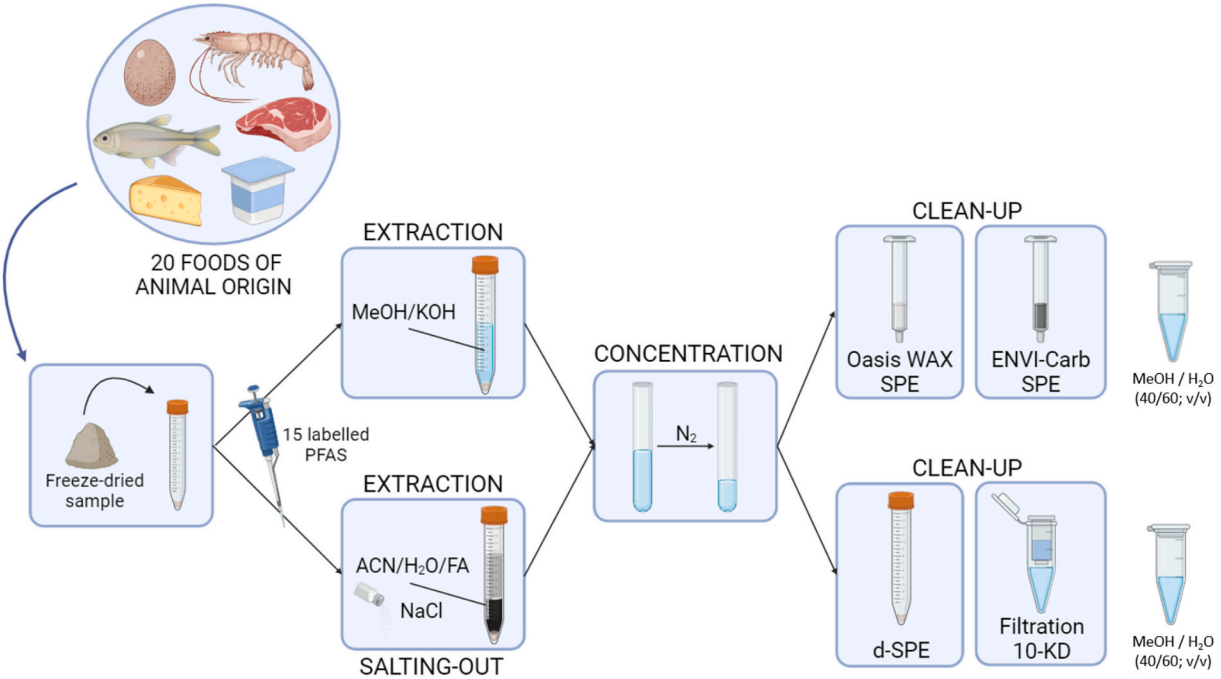


Fig. 1. Detailed sample preparation steps using SPE and QuEChERS methods.

Section 3.1. For each sample, the start of the protocol is the same as the SPE protocol up to the extraction solvent (Fig. 1). A volume of 5 mL of ultra-pure water was added, the mixture was mixed and allowed to soak for 15 min. To these samples, 5 mL of ACN and 50 µL of FA were added, the mixture was mixed, and 600 mg of NaCl were added. The samples were shaken for 1 min, centrifuged for 10 min at 4000 rpm (4 °C), and stored overnight at −20 °C. The supernatant was evaporated under nitrogen to 1 mL. The cleaned-up step was performed by d-SPE with 150 mg of anh. $MgSO_4$ (150 mg), C_{18} (15 mg) and GCB (7.5 mg). The mixture was shaken for 1 min and then centrifuged for 10 min at 4000 rpm (4 °C). Supernatants were filtered through 0.22 µm nylon syringe filters. The extract was dried under nitrogen and reconstituted in the same mixture solvent as the SPE protocol (method A). A variant of this protocol was also tested by adding a filtration step (method B) using Omega 10 kDa membrane (Nanosep Centrifugal Filter; Ugap) (Fig. 1).

2.4. Instrumental analysis

2.4.1. LC-MS/MS analysis

Extracts obtained by applying SPE and QuEChERS protocols were analyzed using an Agilent HP 1200 coupled to a QqQ 6495c Agilent mass spectrometer. The LC-MSMS conditions were based on (Rivière et al., 2014), with minor modifications. The chromatographic separation was performed by a reverse phase C18-column (Hypersil Gold, 100 mm × 2.1 mm i. d., 1.9 µm particle size) equipped with a pre-filter (UHPLC filter 22,180; Thermo Scientific, Bremen, Germany). The column ovens temperature was set to 50 °C. A gradient elution with water 20 mmol. L^{-1} ammonium acetate (A) and methanol (B) was used. The starting mobile phase composition was 30 % B for 0.5 min. B was linearly increased to 67.5 % in 2.5 min, to 69.5 % in 2 min, to 73.5 % in 2 min, and finally to 95 % in 0.5 min (held for 1 min), before returning to 5 % in 0.5 min and 30 % in 1.5 min (held for 0.5 min). The total cycle time was 11 min. The flow rate was 0.6 mL.min^{-1}. The injection volume was set to 10 µL. Triple quadrupole mass spectrometer fitted with an electrospray ionisation (ESI) source operating in negative mode. ESI source conditions were set, as follows: gas temperature 150 °C, nebulizer 6 bars and gas flow rate 650 L.h^{-1}, sheath gas temperature 350 °C, and flow 12 L. min^{-1}, nozzle voltage 0 V, and capillary voltage 2.0 kV in negative mode. Multiple Reaction Monitoring (MRM) was used as an acquisition mode, for a total time of 8.5 min. For all PFASs (except PFBA and PFPeA), two transitions were required to confirm the identity of the analyte (Genualdi et al., 2022). The transitions followed for each molecule are reported in Table S1. For method development purposes, PFASs were quantified using the most intense transition.

2.4.2. LC-ESI(−)-HRMS analysis

Samples extracted by SPE and QuEChERS protocols were analyzed by an Ultimate 3000 UHPLC pumping system coupled to an Orbitrap Q-Exactive mass spectrometer. The LC-HRMS conditions were adapted and modified from the methods used by (Kaufmann et al., 2022) and (Rivière et al., 2014). The chromatographic separation was performed by a reverse phase C18-column (Hypersil Gold, 100 mm × 2.1 mm i. d., 1.9 µm particle size) equipped with a pre-filter (UHPLC filter 22,180; Thermo Scientific, Bremen, Germany). The column ovens and the sample trays temperatures were set to 40 °C and 4 °C, respectively. A gradient elution with water 5 mmol.L^{-1} ammonium formate (A) and acetonitrile (B) was used. The gradient started with 5 % B for 0.5 min, increased linearly to 100 % in 7.5 min (held for 6 min), before returning to 5 % in 4 min (held for 2 min). The total cycle time was 20 min. The flow rate was 0.4 mL.min^{-1}. The injection volume was set to 5 µL. The Q-Exactive mass spectrometer (Thermo Scientific®, Bremen, Germany) fitted with ESI source operating in negative mode. Heated ESI source conditions were set, as follows: sheath gas flow = 55 arbitrary units (AU), auxiliary gas flow rate = 10 AU, auxiliary gas temperature = 300 °C, capillary temperature = 350 °C, spray voltage = 3 kV, and s-lens radio frequency = 40 AU. Data acquisition was performed by full scan and DIA modes. A MS1 full scan mode over the range 65–975 *m/z* at a resolving power of 70,000 at m/z 200 (acquisition window = 0–14 min) operating for the acquisition of the raw data. The automatic gain control (AGC target) and the maximum injection time (IT) were set at 1.10^6 and 100 ms, respectively. After a first processing, a list of suspect compounds is generated, as an inclusion list for non-targeted analyses using DIA mode. This mode used a mass-resolving power of 35,000, with a preselection of the ions to be fragmented as a priority using a precursor ion list (PIL) of the targeted compound, including the retention time and the mass-to-charge ratio. A stepped normalized collision energy of 10, 30, and 90 electronvolts (eV) was employed. The AGC target and the maximum IT were set at 2.10^5 and set to auto, respectively.

2.5. Data processing

For targeted analyses (SPE + LC-MS/MS), Mass Hunter was used for peak integration of LC-MS/MS data. For each compound, two Selected Reaction Monitoring (SRM) (1 quantifier and 1 qualifier) were monitored. The acquisition parameters (SRM transitions and CE voltage) used in LC-MS/MS, in negative electrospray ionisation, for PFASs compounds are presented in Table S1. Labelled internal standards were used to quantify the 28 PFASs according to the isotopic dilution strategy (Rivière et al., 2014). Data representation was performed using R software (version 4.0.1.0).

For the suspect screening analyses (QuEChERS + LC-HRMS), Skyline (Adams et al., 2020; MacLean et al., 2010), a user-friendly open-source software package, was used for peak integration of LC-HRMS data. Peak assignment was made possible by querying an in-house database of available PFASs standards.

Non-targeted screening data analyses (QuEChERS + LC-HRMS) were performed using CD 3.3 SP2 software (Thermo Fishier Scientific) processing raw files acquired in full scan and DIA mode to annotate candidate compounds. The workflow and software parameters used were based on those previously reported, without modification. (Kaufmann et al., 2022; Sanchez & Tautenhahn, 2023).

3. Results and discussion

The objective of this work was to develop an alternative extraction method for PFASs, replacing the routinely used SPE method, and to apply it in both suspect screening (SS) and non-targeted screening (NTS) workflows to evaluate its ability to broaden the range of PFASs detected. As part of this strategy, the approach also involved shifting from conventional targeted MS/MS detection to a more comprehensive HRMS-based strategy, allowing the identification of both known and previously unlisted PFASs in food.

The first step involved developing and optimizing a QuEChERS method for this purpose.

3.1. Optimisation of the QuEChERS extraction and purification steps

To develop a less selective preparation protocol for PFASs analysis in food, the choice was made as an alternative to the SPE towards a QuEChERS. The impact of various parameters was studied as described below and then optimized in order to maximize extraction yields. At this stage, LC-MS/MS was chosen as a reference monitoring method in targeted mode to compare the performance of the tests carried out (§2.4.1). Infant milk and a range of fish from lean to fatty types were selected as relevant matrices for the initial method development. These complex matrices were chosen because they have distinct properties, allowing for a comprehensive assessment of PFASs in different food types.

3.1.1. Extraction solvent

First, the selection of an appropriate solvent is a key determinant of extraction efficiency, thus necessitating the evaluation of multiple solvent options to identify the most suitable choice for the target matrix

and analytes. This approach was first applied to infant milk, followed by fish, to optimize the extraction process for each matrix.

Milk samples were spiked with both the labelled PFASs mix (10 ng/mL, i.e. 0.067 ng/mg milk powder) and the PFASs mix (10 ng/mL). Based on literature on PFASs extraction, the samples were then extracted using five solvent mixtures commonly employed in extraction protocols, as follows: ACN/H_2O (6 mL; 5/1, *v*/v), ACN with 1 % FA (5 mL), ACN/H_2O with 1 % FA (6 mL; 5/1, v/v), MeOH (5 mL), and MTBE/H_2O (6 mL; 5/1, v/v).

The ACN/H_2O/1 % FA mixture demonstrated superior extraction efficiency for PFASs from milk, as indicated by higher signal intensities (Fig. 2) and signal-to-noise (S/N) ratios (Table S2), consistent with previous findings in similar food matrices (Genualdi et al., 2022). This mixture also resulted in low background noise (Table S2), highlighting its efficiency. In comparison, the ACN/H_2O mixture provided equivalent signal intensity for non-polar PFASs (those with retention times exceeding 5 min) but failed to extract some more polar PFASs, those exhibiting shorter retention times, i.e. PFOA, PFHxA, and PFBS, that were either not detected (Fig. 2) or displayed very low S/N ratios (Table S2). Conversely, the ACN/FA mixture was selective for polar PFASs (Fig. 2), emphasizing the complementary nature of the ACN/H_2O/FA extraction method. This result highlights the ability of this mixture to simultaneously extract polar and apolar PFASs, making it a balanced and versatile choice, in line with findings reported in total diet studies (Genualdi et al., 2022). Among the other solvents tested, MeOH only extracted apolar PFASs (Fig. 2), similar to ACN/H_2O, but with lower S/N ratios (Table S2). The MTBE/H_2O solvent allowed extraction of all targeted PFASs (Fig. 2), but the S/N ratioswere lower than those obtained with the ACN/H_2O/FA mixture (Table S2).

The highest concentrations of 4 PFASs measured in the samples, calculated relative to the added internal standards (Fig. S1), suggest that the ACN/H_2O/FA mixture provides superior detection sensitivity and better yields for extracting a wide range of PFASs from milk. This mixture's effectiveness is further supported by its widespread use in various food matrices beyond milk, as evidenced in a total diet study, demonstrating its versatility and reliability in extracting PFASs from diverse food types (Genualdi et al., 2022). This is particularly important in the context of current regulatory frameworks, where achieving high sensitivity in detection is essential, especially when dealing with low-concentration compounds.

Based on these results, the ACN/H2O/FA mixture was retained for further optimization, providing a consistent basis for improving extraction and purification steps.

As for the fish matrix, a large number of protocols in the literature report the extraction of PFASs using a MeOH/KOH mixture (Rivière et al., 2014). Consequently, the ACN/H_2O with 1 % FA (6 mL; 5/1, *v*/v) mixture previoulsy selected for extracting milk was compared to MeOH/KOH (0.01 M; 5 mL), in different fatty and lean fish mixtures. Samples were spiked with both the labelled (1.7 ng.g^{-1} w.w) and the PFASs mix (1.7 ng.g^{-1} w.w). The results (Table S2) show that 90 % of the targeted PFASs exhibit improved S/N ratios, ranging from 2 to 13 times higher, when extracted using ACN/H_2O/FA compared to MeOH/KOH. However, there were some exceptions, such as PFOA for which the S/N ratio decreased by a factor of 3.

Based on the results presented above, for both milk and fish samples, the best extraction conditions were achieved with ACN/H_2O/FA. This method not only efficiently extracts all 28 targeted PFASs, but its ability to accommodate a wide range of physicochemical properties suggests that it has the potential to capture additional, potentially unknown PFASs present in food items, thus expanding the chemical space that can be extracted.

3.1.2. *Extraction salts*

In QuEChERS methods, two main salt compositions are commonly reported in the literature for the extraction step (Genualdi et al., 2022; Roberts et al., 2023). Their performance was compared to determine which provides the highest signal-to-noise (S/N) ratios, specifically using either a mixture of $MgSO_4$/NaCl (600 mg; 4:1, *w*/w) or NaCl alone (600 mg), as detailed in Table S2. The addition of NaCl alone allows the extraction of over 80 % of the 28 PFASs targeted with a S/N ratio up to

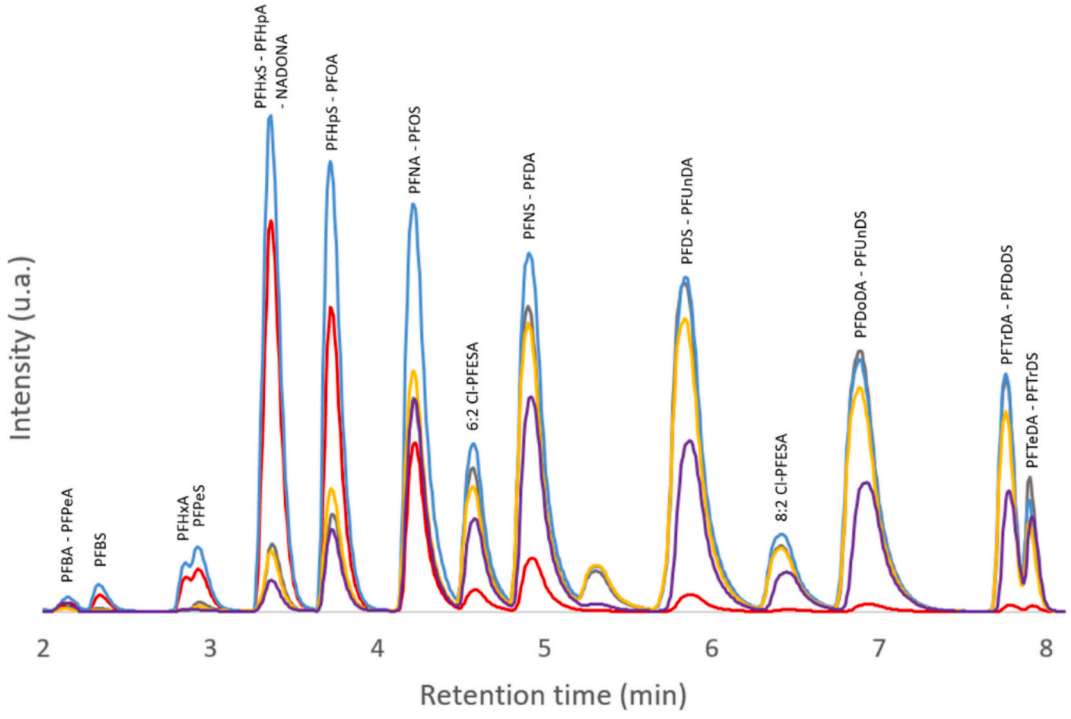


Fig. 2. Relative signal Intensities (TICs) observed by LC-(ESI-)MS/MS for the analysis of PFASs in spiked milk (10 ng/mL) extracted by QuEChERS comparing 5 different solvent mixtures: ACN/H_2O (grey), ACN/FA (red), ACN/H_2O/FA (blue), MeOH (yellow) et MTBE/H_2O (purple). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2.5 times higher. The $MgSO_4$/NaCl mixture did not provide better results, except in rare cases where the S/N ratio was similar or slightly greater than with NaCl alone (e.g. for PFOS or PFTrDS, with S/N 1.1 and 1.5 times greater, resp.) (Table S2). This study did not demonstrate $MgSO_4$ was essential in the salting-out step, unlike what is often reported in the literature. As a result, adding only NaCl to this step was selected as a fast and efficient method.

Whereas literature protocols indicate that the extraction process is generally performed at room temperature for at least 15 h (Rivière et al., 2014), this parameter was also investigated at 4 °C and − 20 °C to allow a better separation between the organic phase and the aqueous phase. Surprisingly, the results show that this parameter does not have a major impact (Table S2), and was consequently set at −20 °C.

3.1.3. Purification step

Then, because of the complexity of some food matrices, a clean-up step of the extracts is crucial in sample preparation in order to increase detection performance such as sensitivity and selectivity. Although numerous purification methods have been reported in the literature, some methods appear to be more suitable depending on both the matrice and the selected PFASs (Table 1). To eliminate interferents while preserving the analytes of interest, the second part of the protocol QuEChERS (d-SPE step) must therefore resolve this difficult stage. Consequently, $MgSO_4$/C_{18}/GCB was selected for the d-SPE step as it is considered in the literature to be the best performing and most frequently used (Genualdi et al., 2022; Lacina et al., 2011; Taylor & Sapozhnikova, 2022). Additionnaly, dispersive liquid–liquid microextraction (DLLME), using 10 % NaCl solution and chloroform, which is used in the reference method, was tested. The DLLME method is known to increase the size of the extraction solvent contact surface, resulting in a higher enrichment factor (Leong et al., 2014). Finally, and because of the risk of eliminating compounds of interest during this purification step, the performance of an extraction method without purification on increasing S/N ratios should also be compared. Because this crucial step also heavily depends on the matrix properties, those three conditions have been tested in various fish an milk and the outcome presented in Fig. S2.

The same trends in results were observed in these very different matrices (Fig. S2). Indeed, the DLLME purification allows detection with the highest S/N ratios of apolar PFASs. However, d-SPE step or no cleaning enabled extracting more polar PFASs, with S/N ratios up to 4 times higher using d-SPE, than the DLLME method. The DLLME protocol was further investigated, highlighting the presence of several polar PFASs in the unprocessed aqueous phase, and was therefore unsuitable to extract all targeted PFASs. These results clearly illustrate the difficulty of finding compromises when addressing compounds with such diverse physico-chemical properties. Finally, the d-SPE method was selected as the best compromise for the purification step for both milk and fish samples (Fig. S2).

As an additional lever to increase the detection sensitivity of analytes, the volume of recovery solvent, was therefore reduced from 130 µL to 90 µL, allowing a reasonable concentration of analytes without altering the quality of the signal.

3.1.4. Additional purification step

Finally, optimisation of the various QuEChERS parameters enabled optimal extraction of the 28 PFASs initially targeted in the study. Although a slight white deposit, probably salt due to the use of NaCl without $MgSO_4$, was observed on the cone of the LC-MS/MS instrument, the quantification of the analytes was not affected. In order to move towards a more global and less targeted workflow, these extracts were then analyzed by LC-HRMS to anticipate future developments in this work. Clogging of the transfer capillary was observed, resulting in the implementation of a further purification step, as already reported as necessary in the literature, with all the challenges associated with adding such a step which must not compromise PFASs recovery (Roberts et al., 2023).

Although this desalting step can be performed using SPE, alternative options were explored in order to preserve the simplicity and rapidity of the QuEChERS protocol. Consequently, two purification steps were assessed, alone or in combination, to address this crucial step: a PD-10 desalting column and a Omega 10 KDa membrane which performances were compared to the previously optimsed QuEChERS method, which does not involve any additional purification step (Table S2).

The extracts desalted using PD-10 were filtered through a 10 kDa cut-off membrane filter due to their opacity, and the quality of the signal was not deemed acceptable, based on an S/N ratio reduction of up to 5 times, compared with the optimized method. An additional purification step using a 10 kDa centrifugal filter improved signal clarity by reducing background noise, leading to up to a threefold increase in the signal-to-noise ratio for more than 75 % of the PFASs (Table S2). This filter was chosen to minimize matrix interferences while preserving both the analytical equipment and its performance.

3.1.5. Comparative analysis of two sample preparation protocols for food analysis

The study then focused on applying both the SPE method, which serves as the reference protocol, and the optimized QuEChERS protocol to twenty food samples of animal origin, representative of typical consumption in France. Each PFAS was spiked into the samples at concentrations of 1, 5, and 10 $\mu g \cdot kg^{-1}$ (wet weight), in accordance with the concentration levels established for fish in Regulation 2022/2388, specifically for the four compounds PFOS, PFOA, PFNA, and PFHxS (European Commission, 2022a). To compare the performance of the two protocols, the four regulated PFASs were quantified in 20 matrices using both the SPE and QuEChERS sample preparation methods, coupled with LC-MS/MS analysis. The concentrations of each PFAS at 1, 5, and 10 $\mu g \cdot kg^{-1}$ were determined, and the mean and standard deviation for each level were calculated. The results presented in Fig. 3 confirm the capabilities of both protocols to deliver quantitative results although a slight dispersion of the quantified values is observed as the concentration of PFASs increases, especially for the QuEChERS method. As illustrated in Fig. S3, this dispersion is strongly dependent on the matrix studied. For example, a slight dispersion is observed in crustaceans, unlike in fish, and the same trends were observed for the other PFASs. These results demonstrate the suitability of the QuEChERS protocol to deliver quantitative results for the 28 PFASs in various matrices of animal origin. The next step is to apply this generic sample preparation in a non-targeted workflow involving HRMS acquisition, aiming to evaluate the potential for expanding the chemical space of PFASs accessible for analysis.

3.2. Application of the QuEChERS protocol to the search for the presence of additional PFASs

The same twenty food samples of animal origin (3 eggs, 4 molluscs and crustaceans, 5 fish, 5 meats, and 3 dairy products) used for method development, where spiking was necessary, were extracted using the optimized QuEChERS protocol without any spiked compounds. This step aimed to investigate the ‘native’ PFASs contamination of these samples. In order to continue the comparison between the two extraction protocols to the end, these 20 samples were also extracted by SPE.

3.2.1. Suspect screening (SS)

The 20 extracts obtained after applying both protocols were analyzed by LC-HRMS and the acquired data first processed by Skyline software. Identification was performed according to an in-house database, including the 28 PFASs.

The characterization of the 20 food items using the suspect screening (SS) approach, i.e. based on QuEChERS extraction and LC-HRMS analysis, led to the establishment of specific contamination profiles for each food category. In total, 2 PFASs (PFBA and PFOA) were detected in meat and dairy samples, 4 in fish (PFBA, PFPeA, PFHpA and PFOA) and

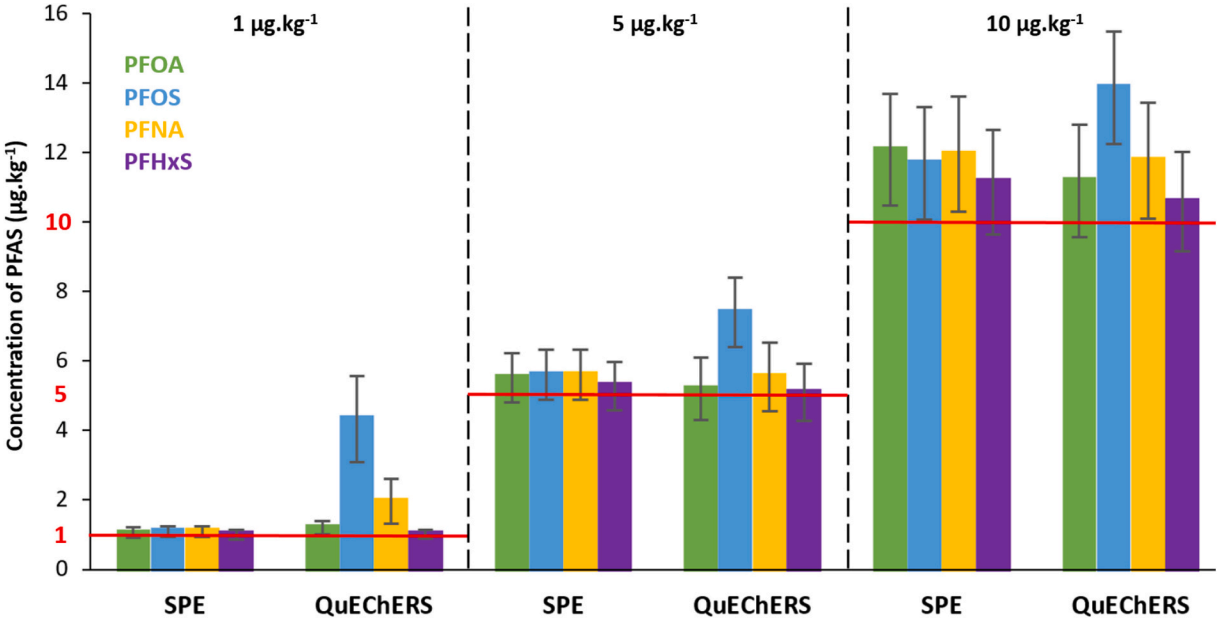


Fig. 3. Concentrations of 4 selected PFASs (PFOA, PFOS, PFNA, PFHxS) measured by LC-MS/MS following the application of either the SPE or QuEChERS extraction protocol, with supplementation at three levels (1, 5, and 10 µg·kg^{-1}, wet weight) across 20 food items.

shellfish (PFBA, PFPeA, PFHxA and PFOA), and up to 6 (PFBA, PFPeA, PFHxA, PFHpA, PFOA and PFNA) in egg samples. These findings are consistent with existing occurrence data reported in the literature, confirming the higher susceptibility of certain food matrices—particularly eggs and aquatic products—to PFAS contamination (EFSA, 2020). This initial screening phase thus provided a solid basis for further investigation by non-targeted analysis. Further, this approach made it possible to detect some PFASs compounds that were

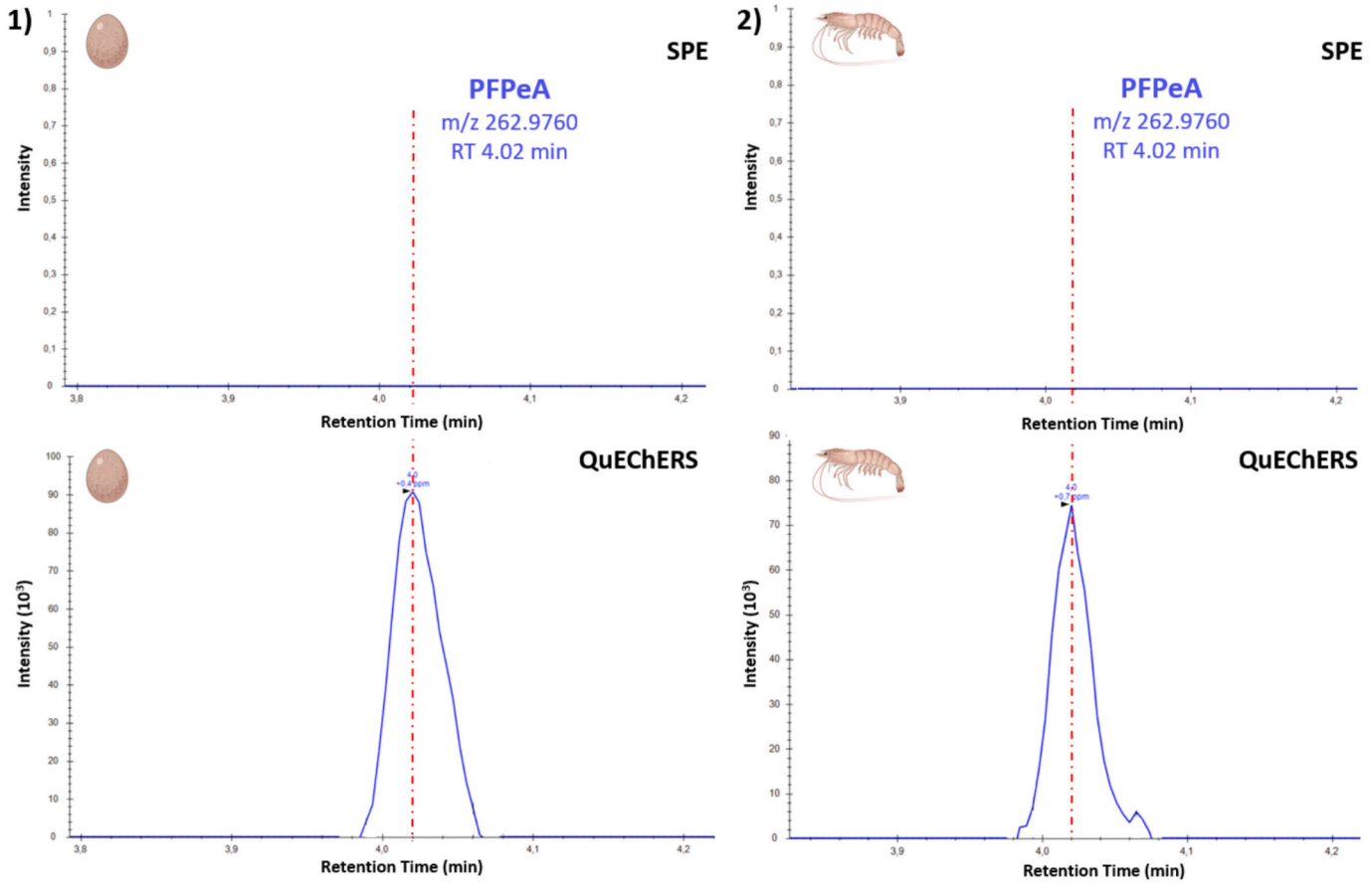


Fig. 4. Chromatographic profiles of PFPeA obtained from (1) an egg sample and (2) a crustacean sample in SS acquisition mode, comparing the extraction protocols: SPE and QuEChERS.

not extracted by the protocol using SPE, such as PFPeA in egg and crustacean samples (Fig. 4), and PFOA, PFHxA, PFHpA and PFBA in eggs, meat and dairy products samples (Fig. S4).

In addition, while QuEChERS demonstrates its broader capacity to extract PFASs compared to SPE, this study also highlights some limitations, particularly for PFASs with higher molecular weights (> 550 g/mol), such as PFNA. These compounds may be adsorbed onto the 10 kDa filter, which has been employed in biological applications (Vergauwen et al., 2017). However, to the best of our knowledge, no previous studies have specifically investigated the impact of this filter on the analysis of PFASs. It thus appears that there is complementarity between QuEChERS, particularly effective for PFASs with Mw <500 g.mol^{-1}, and SPE that would be more suitable for those with higher Mw. As most emerging PFASs are short aliphatic chain (≤ C7) compounds (Capriotti et al., 2013), the QuEChERS protocol developed in the present study shows great potential to address their monitoring and extend the number of relevant PFASs analyzed in foods. Based on these promising results, an NTA approach was initiated, involving specific data processing and prioritization strategy enabling non-targeted search for compounds.

3.2.2. *Non-targeted screening*

To further explore the detection of additional PFASs, a non-targeted data processing strategy was subsequently implemented to prioritise signals to be investigated. Previously acquired LC-HRMS data were processed using Compound Discoverer for automated feature annotation. PFASs prioritization relied on the mass defect associated with fluorine atoms to isolate potential candidates. In this exploratory framework, m/C versus md/C representation and KMD plots (Fig. 5) proved particularly useful, as they allow the visualization of structurally related compounds with varying masses, facilitating the identification of novel series of PFASs homologues (Joerss & Menger, 2023; Kaufmann et al., 2022).

The application of the NTS workflow to the different foodstuffs confirmed the suitability of the overall strategy (QuEChERS + HRMS) by successfully detecting PFASs compounds already identified through targeted or SS (e.g., PFOA) approaches (Fig. 5a).

More importantly, this approach, through the data processing and signal prioritization strategy implemented, led to the detection of a previously unlisted compound (Fig. 5a). Mass defect plots have thus been used to determine PFASs homologous series. In this case, the masses of the 28 PFASs initially involved in the study were normalized to CF2 leading to the normalized mass defect plot in Fig. 5a. Visualization in a CF2-based Kendrick mass defect plot easily allows a first overview on the data since compounds belonging to the same class are aligned horizontally. Most of the 28 PFASs could thus be observed across 3 main homologous series based on similar normalized mass defect for a series and mass differences divisible by 49.9968 (CF2). By applying this strategy to the food samples studied in this study, an unknown signal aligned with the perfluoroalkyl carboxylic acids (PFCA) series was detected in an egg (Fig. 5a). Specific study of this signal, in particular the *m/z* and the associated fragmentation profile, led to the hypothesis that it was perfluoropropanoic acid (PFPrA) (Fig. S5). Since this compound was not included in the reference list of the targeted method or our database, it remained undetected in both the targeted and SS approaches. To validate this finding an analytical standard of PFPrA was acquired, allowing its identification to be assigned a confidence level of 1a (Fig. 5b and c), in accordance with current reference standards in this field (Charbonnet et al. 2022). This study not only presents the first known detection of Perfluoropropanoic acid (PFPrA) in food matrices—specifically in eggs—but also introduces an innovative QuEChERS-based extraction methodology coupled with HRMS for broader PFASs screening. This combined methodological and application advancement significantly expands the chemical space accessible for PFAS analysis in food, offering novel tools to address emerging contaminants.

Although PFPrA has been previously identified in environmental matrices such as both potable (treated) and non-potable water sources

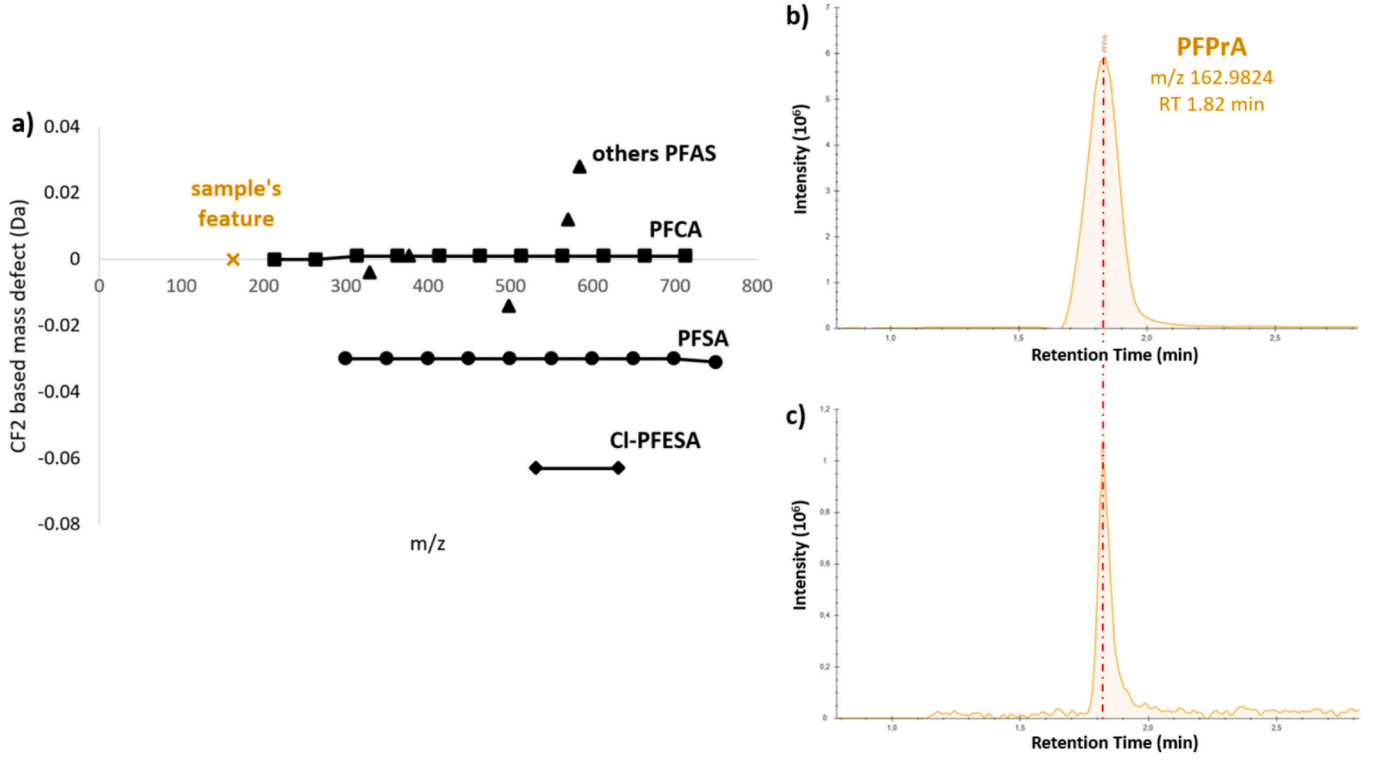


Fig. 5. a) Kendrick mass defect for CF2 repeating unit vs *m/z* of 28 PFASs standard listed in our internal database and classified by family: PFCA ($n = 11$), PFSA ($n = 10$), Cl-PFESA ($n = 2$) and others PFASs ($n = 5$). Homologous series are identified by shifts in the x-axis divisible by 49.9968 (CF2) and the same CF2 normalized mass defect. In yellow, a suspect PFASs signal highlighted by NTS data processing. b) Chromatogram profile for PFPrA standard. c) Chromatogram profile for PFPrA in an egg sample. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(Lenka et al., 2022; Liang et al., 2023), its presence in foodstuffs has not been reported until now. This finding suggests the potential transfer of PFPrA through the food chain, highlighting the need for further investigation into its bioaccumulation and associated health risks, particularly since its detection in foodstuffs has not been previously reported.

3.3. Limitations and future directions

- The strategy involved developing a sample preparation method optimized for known PFASs ($n = 28$) and then applying it to uncover previously unmonitored compounds. This approach successfully led to the identification of perfluoropropanoic acid (PFPrA), a PFAS not typically extracted using standard methods such as SPE. However, one limitation of this strategy is that the method was initially designed based on a selected subset of known PFASs, primarily those regulated or commonly monitored. As a result, it may not have been fully optimized for extracting other PFASs with significantly different physicochemical properties, such as ultra-short-chain PFASs or highly functionalized derivatives. This constraint could have impacted the detection of additional unknown compounds, highlighting the need for further refinement of extraction conditions to encompass an even broader range of PFASs.
- At the end of the sample preparation protocol, an additional purification step was necessary to ensure the removal of residual impurities. A desalting column and an additional filter were therefore tested. However, studies such as (Taylor & Sapozhnikova, 2022) have highlighted ultracentrifugation as a highly effective purification method. Given its potential to improve the overall process, evaluating ultracentrifugation could be a valuable next step. This approach will be considered in future studies to assess its ability to further enhance the purity and sensitivity of the final samples for PFASs detection.
- Regarding data acquisition, while Data-Dependent Acquisition (DDA) mode is ideal for focusing on the most abundant ions, technical limitations necessitated the use of Data-Independent Acquisition (DIA) mode in our study. DIA however provides comprehensive data by fragmenting all precursor ions across the chromatogram, increasing the likelihood of identifying unknown compounds, including novel PFASs.

4. Conclusion

This study aimed to develop a suspect and non-targeted approach to identify PFASs beyond those traditionally monitored in foodstuffs. To achieve this challenging objective, several complementary steps were undertaken, including sample preparation, analytical method development, and data processing. Sample preparation was optimized to address the challenge of extracting PFASs from complex food matrices at trace levels. Specifically, a broader extraction method, based on the QuEChERS approach and supported by literature, was evaluated. Multiple parameters known to significantly influence the extraction and detection of PFASs were investigated and optimized. Focusing on 28 targeted PFASs in a selection of 20 commercial food samples, the results demonstrated that the optimized QuEChERS protocol enabled the detection of PFASs that were not extractable using the current SPE-based method.

In parallel, the study successfully applied, for one of the first times, a non-targeted screening (NTS) strategy—typically used in environmental contexts—to the analysis of foodstuffs. By combining the optimized QuEChERS extraction with HRMS acquisition and advanced signal prioritization workflows, the approach extended the detection scope beyond conventional targeted methods. This enabled the identification of both known and previously unlisted PFASs, including the detection of perfluoropropanoic acid (PFPrA) in an egg sample, underscoring the method's capacity to reveal overlooked contaminants in the food chain.

These findings open new perspectives for PFAS surveillance and regulatory control in food, supporting the implementation of HRMS-based strategies for risk assessment. They highlight the potential for better characterization and mitigation of consumer exposure to persistent contaminants. Future research should focus on large-scale and geographically diverse sampling to refine exposure assessment. A particular emphasis on samples with high levels of monitored PFASs may facilitate the identification of additional, less-studied compounds. Furthermore, integrating HRMS-based workflows into routine food monitoring could enhance the development of adaptive regulatory frameworks aligned with the evolving challenges of PFAS contamination.

CRediT authorship contribution statement

Cassandre Jeannot: Writing – review & editing, Writing – original draft, Investigation, Data curation. **Nicolas Macorps:** Investigation. **Bruno Le Bizec:** Funding acquisition, Conceptualization. **Julien Parinet:** Supervision, Project administration, Funding acquisition, Conceptualization. **Gaud Dervilly:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fochx.2025.102674.

Data availability

Data will be made available on request.

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